Literature Review: Deep Learning and AI-Based Cattle Health Monitoring (2026 Studies)

Compiled from article pages and abstracts. Limitations are my assessment from those sources; verify against the full text before citing. Items marked "not confirmed" or "not retrieved" could not be verified. Ref 5 is a preprint; Refs 9 and 10 are reviews.

Ref	Paper Title	Author(s) / Year	Journal	Month & Year	Methodology & Technology	Advantages	Limitations
1	Empowering Rural Livestock Health: AI-Powered Early Detection of Cattle Diseases	Rao, D. S., Reddy, P. C. S., Revathi, A. et al. (2026)	AI, 7(4), 137	Apr 2026	Mobile app (React Native) collects cattle images and farm data into an Azure cloud database. Five ImageNet-pretrained CNNs (EfficientNetB0, EfficientNetV2B0/S, ResNet50, VGG16) were fine-tuned for Lumpy Skin Disease (binary) and Foot and Mouth Disease (4-class), with a soft-voting ensemble and a Streamlit/Folium geospatial dashboard.	End-to-end field pipeline (capture, cloud inference, dashboard) aimed at rural veterinary use. LSD test accuracy 98.36% (EfficientNetB0); FMD ensemble accuracy 99.84%. Split at animal level to avoid leakage.	Internal validation only. Small FMD set (210 images per class) and a large train-test gap (e.g., 99.47% vs 94.05%), which the authors say may reflect memorization. Ensemble was not compared with single models. Detects visible lesions, not pre-clinical disease.
2	Multi-scale Keypoints Detection and Motion Features Extraction in Dairy Cows Using ResNet101-ASPP Network	Wu, S., Han, S., Zhang, J. et al. (2026)	Journal of Integrative Agriculture, 25(5), 2028–2040	May 2026	ResNet101 backbone with an atrous spatial pyramid pooling (ASPP) module for multi-scale keypoint detection; seven motion features (gait symmetry, step-height balance, head swing, head-neck slope, back curvature, etc.) extracted from tracked keypoints and compared across healthy, mildly lame and severely lame cows (2,385-frame dataset).	Handles scale changes and occlusion; prediction confidence rose 1.65–2.50% across three cow scales. All seven features helped separate lameness grades. Model is reusable for behavior and estrus tasks.	Small dataset (2,385 frames). Gain over the baseline is small (about 0.24–0.27 px error). Abstract reports feature distributions, not a lameness classifier accuracy. Dairy cows only; number of farms not confirmed.
3	Infrared Thermography and Machine Learning for Mastitis Detection in Dairy Cows: A Pilot Case Study in Egyptian Farms	Elmasry, A. S., Elwakeel, E. A. et al. (2026)	Veterinary Sciences, 13(7), 640	2026 (month not confirmed)	976 thermal udder images from 488 Holstein cows (708 healthy, 268 mastitic) on two farms. EfficientNetB3 with global average and max pooling gave 3,072 features; ten classifiers were compared. Also explored links to somatic cell count, blood biomarkers and feed ration.	Non-invasive screening. Best image-level hold-out result (MLP): accuracy 86.22%, AUC 0.918, specificity 90.85%. Reports a leakage-aware, cow-grouped validation alongside the optimistic one.	Sensitivity only 74.07%. Group-based 5-fold CV dropped to AUC 0.68 and accuracy 0.62. No significant association with SCC, CMT, biomarkers or nutrition. Two farms; pilot study.
4	Thermal Image-Based Artificial Neural Network Approach to Determine Mastitis Detection in Holstein Dairy Cattle	Şahin, H. A., Mikó, E., Önder, H. & Baccouri, W. (2026)	Animals, 16(7), 1048	2026 (month not confirmed)	ANN trained on RGB thermal images of the udder taken at milking from 500 Holstein cows. Mastitis levels were set from California Mastitis Test scores with somatic cell count as the output variable; data split 70/15/15.	Large sample (500 cows) gathered during routine milking; non-invasive. Authors conclude ANN plus thermal imaging gives reliable results.	I could not retrieve the performance figures. Single breed. Not confirmed whether the split was by cow or by image. Shallow ANN, not a modern CNN.
5	Can 3D Point Cloud Data Improve Automated Body Condition Score Prediction in Dairy Cattle?	Tang, Z. et al. (2026)	arXiv preprint 2601.22522	Jan 2026	Top-view depth images were compared with 3D point clouds for body condition score (BCS) prediction under four settings (raw, segmented full body, segmented hindquarter, handcrafted features) using data from 1,020 cows on a commercial farm, with cow-level cross-validation.	Fair head-to-head comparison with leakage control. Shows depth images matched or beat point clouds, so cheaper sensors may be enough. BCS links to metabolic status and health.	Preprint; peer-review status not confirmed. Single commercial farm. I did not retrieve numeric accuracy. Point clouds were noise-sensitive; handcrafted features lowered accuracy.

Ref	Paper Title	Author(s) / Year	Journal	Month & Year	Methodology & Technology	Advantages	Limitations
6	Non-Contact Respiration Rate Estimation in Cattle from Overhead Videos	Bhattacharya, P., Endlicher, E., Ravinaidu, G., Bieber, G. et al. (2026)	VISAPP 2026, Vol. 1, pp. 582–589	2026 (month not confirmed)	Overhead RGB camera; Segment Anything Model (SAM) segments the animal, area or contour features give a time signal, then band-pass filtering (10–100 breaths/min) and peak picking estimate respiration rate. Tested on 9 video snippets from two German farms against counts by five observers.	No wearable needed; overhead view reduces occlusion. Mean absolute error 2.11 BPM (partial mask), about 5% at worst across four feature options. Works for cows and calves.	Semi-automated: manual ROI, sub-ROI and choice of still, unoccluded clips. Only nine snippets. Weak with vigorous movement or unstable cameras. Human counts, not a sensor, as reference; no disease validation.
7	Non-Contact Detection of Post-Regurgitation Deep Inhalation in Calves Using Infrared Thermography and Deep Learning-Based Nostril Segmentation	Kim, S. et al. (2026; author list not confirmed)	BMC Veterinary (journal name as given in press release; not confirmed)	Feb 2026	Synchronized RGB and thermal video of 8 calves over 28 sessions. Deep learning segments the nostril in each RGB frame to read temperature and build breathing patterns, which are used to tell post-regurgitation deep inhalation from other inhalations.	Fully non-contact rumination-linked signal, with no attached sensor. Links breathing physiology to rumination, a health indicator.	Described by the authors as proof of concept. Only 8 calves. I did not retrieve accuracy figures. Measures a rumination event, not disease directly.
8	Smart Livestock: A Federated Learning Based System for Real-Time Cattle Health, Stress and Water Resource Monitoring	Gulati, V., Grover, R., Kumar, N., Jindal, P. & Shaheen, M. (2026)	Veterinary Medicine and Science, 12(3), e70979	May 2026	Federated learning with a CNN-LSTM trained locally on edge devices; evaluated on the MmCows multimodal dataset (wearable physiological, behavioral and environmental data) for health and stress classification, compared with centralized models.	Raw sensor data stay on each farm (privacy); suited to low-connectivity rural settings. Reported F1 of about 0.91 in the abstract snippet I retrieved.	Tested on a public dataset (abstract mentions real or simulated data); field deployment not confirmed. Needs wearables on each animal. Full accuracy figures not retrieved.
9	Computer Vision for Cattle Health and Welfare Monitoring: A Comprehensive Review of Methods, Applications, and Interdisciplinary Integration in Smart Agriculture	Islam, M. N., Edwards, J. L., Burns, R., Qi, H. & Gan, H. (2026)	Sensors, 26(13), 4271	Jul 2026	Systematic review of 140 studies (Jan 2015–May 2026) from six databases covering body condition, lameness, weight, estrus, feeding/drinking, breathing and behavior, with sensor and camera-placement guidance.	Broad, practical coverage. Reports accuracy ranges (e.g., lameness 71.9–98.89%; breathing 87–96.8%). Highlights non-invasive, scalable monitoring and edge deployment potential.	Review only (no new experiments). Most included studies are on dairy cows. Authors note environmental variability, large annotation needs and imaging cost; disease-specific monitoring is still underexplored.
10	A Review on the Impact of AI-Enabled Thermal Imaging and IoT Sensor Fusion on Early Detection of Mastitis in Dairy Cattle	Asogan, A., Sazali, N., Veerendra, A. S., Samylingam, L., Aslfattahi, N., Kok, C. K. & Kadirgama, K. (2026)	Biosensors and Bioelectronics: X, 28, 100735	Mar 2026	Narrative review of AI methods, infrared thermography and IoT devices (milking systems, behavioral collars) for early mastitis detection, set against clinical inspection, CMT and somatic cell count.	Focuses on early (subclinical) detection and on combining image and sensor data; discusses cost-effective herd-level use.	Review only; I read the abstract, not the full text. Abstract states AI analytics with thermography and IoT devices have yet to be integrated, so fusion evidence is limited.

Research Gap

Synthesizing the limitations reported across the ten reviewed studies (2026), the following gaps emerge in AI-based cattle health monitoring:

- **Single-condition silos; disease-specific monitoring is thin.** Studies target one problem at a time: LSD/FMD (Ref 1), lameness (Ref 2), mastitis (Refs 3, 4, 10), body condition (Ref 5) or breathing (Refs 6, 7). Ref 9 itself flags disease-specific monitoring as underexplored. No study combines several health conditions in one system.
- **Small, single-site, internally validated datasets.** Ref 1 uses internal validation with 210 FMD images per class; Ref 2 uses 2,385 frames; Ref 6 uses nine video snippets; Ref 7 uses eight calves; Ref 3 covers two farms. This limits generalization across breeds, housing and lighting.
- **Optimistic validation hides real-world performance.** Ref 3 falls from AUC 0.918 (image-level hold-out) to 0.68 (cow-grouped CV). Ref 5 uses cow-level CV, which is the stronger practice, but it is rare in the other studies; Ref 4 does not confirm its split unit.
- **Weak link to clinical ground truth.** Ref 3 found no significant association between model predictions and SCC, CMT, blood biomarkers or nutrition. Respiration work (Ref 6) is checked against human counts, not a sensor, and Ref 7 measures a rumination event, not disease.
- **Early or pre-clinical detection is not demonstrated.** Ref 1 recognizes visible lesions in cases that are already symptomatic. No reviewed study follows animals prospectively to show how many days earlier AI flags a problem than farm staff or a vet.
- **Manual steps and no real-time field operation.** Ref 6 needs manually drawn regions and hand-picked still clips; Ref 7 is a proof of concept. Deployed, continuous, unattended systems with farm-level results are missing.
- **Sensor cost and modality trade-offs.** Thermal, depth and 3D cameras raise cost. Ref 5 shows point clouds were more noise-sensitive without a consistent gain over depth images, and Ref 9 lists imaging cost and environmental variability as barriers.
- **Limited fusion of vision, thermal and wearable data.** Vision and thermal studies (Refs 1–7) and sensor-based work (Ref 8) are mostly separate. Ref 10 notes that AI with thermography and IoT devices has yet to be integrated.
- **Privacy-preserving, multi-farm learning is not yet field-proven.** Ref 8 shows federated learning on a public dataset; deployment on real farms, with wearable burden and low connectivity, is not confirmed.
- **No unified benchmark or adoption evidence.** Studies report accuracy, AUC, F1, mean absolute error (BPM) or confidence gain, on different datasets, so results cannot be compared. Few report cost, labor savings, treatment outcomes or usability for smaller farms.

SDG Mapping

The reviewed literature aligns with the following United Nations Sustainable Development Goals (SDGs):

SDG	Relevant Studies	Mapping Rationale
SDG 2 - Zero Hunger	Refs 1–10 (all)	Earlier detection of disease, lameness and mastitis protects livestock productivity and the milk and meat supply.
SDG 3 - Good Health and Well-Being	Refs 1, 3, 4, 10	Early detection of contagious diseases (LSD, FMD) and mastitis supports milk quality and lowers the risk of disease reaching people through milk and animal products (Ref 1 raises this concern).
SDG 6 - Clean Water and Sanitation	Ref 8	Ref 8 pairs cattle health monitoring with water resource monitoring on farms; water-resource claims were not verified beyond the title and abstract.
SDG 9 - Industry, Innovation and Infrastructure	Refs 1, 2, 5, 6, 8, 9	Edge AI, federated learning, cloud dashboards, depth and thermal imaging are innovations in agricultural infrastructure.
SDG 12 - Responsible Consumption and Production	Refs 3, 4, 5, 9, 10	Catching mastitis and poor body condition early reduces losses in milk output, culling and wasted feed and resources.

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